Mrs. Payalben H Hinsu | A.Y.: 2025-26 | Sem./Year: 3 | Course: 2302CS303 - Database Management System - II | Slot Type: Lab (TA) | Division: CE-3A

Sr.	Course Content	Planning Title	Planning Description
1	Introduction of T – SQL and Database objects	Lab-1	Create the following tables with apply constraints. 1) Table No.1 – Bank_Master (Bank_ID (PK), Bank_Name, Bank_ShortName) 2) Table No.2 – Branch_Master (Branch_ID (PK), Branch_Name, Branch_IFSC (Unique), Bank_ID (FK)) 3) Table No.3 – Employee_Master (Emp_No (PK), Branch_IFSC (FK), Emp_FullName, Emp_Designation, Emp_Manager_No, Employee_Salary) 4) Table No.4 – Customer_Master (Cust_ID (PK), Cust_FullName, Cust_DOB, Cust_Address, Cust_MobileNo, Cust_EmailID, Cust_City) 5) Table No.5 – Account_Master (Acc_No (PK), Cust_ID (FK), Acc_Type (Must Contain only values SB -> Saving, CR -> Current), Branch_IFSC(FK)) 6) Table No.6 – Transaction_Master (Tran_ID (PK), Tras_Acc_No (FK), Tran_Date, Tran_Type (Must Contain only values CH -> Cash, CQ -> Cheque, OL -> Online, RG -> RTGS), Tran_Amount_Debit_Credit (Must Contain only values D -> Debit, C -> Credit), Tran_Amount)
2	Introduction of T – SQL and Database objects	Lab-2	Insert records into the table that we created in Practical 1.
3	Introduction of T – SQL and Database objects	Lab-3	1) Display Employee number, name and branch name. **(A)** 2) Display Account number, customer id, name and branch IFSC code using join. **(A)** 3) Display Transaction ID, amount, account number, account type whose transaction type is Online. **(A)** 4) Display Account number, type, transaction account number and amount using left outer join. **(A)** 5) Display Account number, type, transaction account number and amount using right outer join. **(A)** 6) Display customer name, mobile number who has highest transaction amount. **(A)** 7) Display Branch name, IFSC and Bank ID who has lowest paying amount employee. **(A)** 8) Display the count of total designation of an employees. **(A)** 9) Display the count of how many customers have saving account. **(A)** 10) Display details of branch master branch name wise in descending order. **(A)**
4	Introduction of T – SQL and Database objects	Lab-4	1) Write a T - SQL block to check whether the given number is a positive number or a negative number using a simple if statement. **(A)** 2) Write a T - SQL block to find the maximum number from the given two numbers. **(A)** 3) Write a T - SQL block to find the maximum number from the given three numbers. **(A)** 4) Write a T - SQL block to print the first 25 natural numbers using a loop. **(A)** 5) Write a T - SQL Program to Print Odd Numbers From 1 to 100. **(A)** 6) Write a T - SQL block to find the sum of the first 100 natural nos. **(B)** 7) Write a T - SQL block to find whether the number is even or odd. **(B)** 8) Write a T - SQL block to print the first 25 Odd numbers using a loop in Reverse order. **(C)** 9) Write a T-SQL block for given conditions: marks > 70 then Print '1st Class', marks>50 and marks<=70 then print '2nd Class', marks>=35 and marks<=50 then print '3rd Class', marks<35 then print 'Fail !!'. **(C)**

Mrs. Payalben H Hinsu | A.Y.: 2025-26 | Sem./Year: 3 | Course: 2302CS303 - Database Management System - II | Slot Type: Lab (TA) | Division: CE-3A

Sr.	Course Content	Planning Title	Planning Description
5	Introduction of T – SQL and Database objects	Lab-5	1) Create a simple view Bank_View which contains only Bank_ID and Bank_Name. **(A)** 2) Create a simple view Customer_View which contains Cust_FullName, Cust_MobileNo, Cust_EmailID. **(A)** 3) Create a complex view that contains Acc_No, Cust_ID, Branch_Name, Bank_Name. **(A)** 4) Create a simple view with Check Option Cust_View which contains Cust_FullName, Cust_City. **(A)** 5) Create a sequence on the following table's listed columns: Bank_Master -> Bank_ID, Branch_Master -> Branch_ID, Employee_Master -> Emp_No. **(A)** 6) Create Synonym T_Master for Transaction_Master. **(A)** 7) Create a simple view Tr_View which contains Tran_ID, Tras_Acc_No those whose Tran_Type is Online. **(B)** 8) Create whole view of the Custome_Master table with check option for customer city = 'Rajkot' and insert data using view. **(B)** 9) Create a one Table and create sequence for which contain start from 101 increment by 5 and maximum value is 120, minimum value is 100 and restart cycle after reach on maximum value. **(B)** 10) Insert one value into the Employee_Master table fetch next id from using sequence. **(B)** 11) Alter Tr_View that also contains Tran_Amount column. **(C)** 12) Drop the Customer_View. **(C)** 13) Drop the Employee_Master table sequence and try to insert data using sequence. **(C)**
6	Function, Stored Procedures, Trigger and Cursor	Lab-6	1) Create a function to calculate simple interest (SI=P*R*N/100). **(A)** 2) Create a function to find the feet to inch. **(A)** 3) Create a function to convert Celsius to Fahrenheit. **(B)** 4) Create a function to find factorial of a number. **(C)** 5) Write a function to find the sum of odd numbers between two range. **(C)**
7	Function, Stored Procedures, Trigger and Cursor	Lab-7	1) Write a stored procedure to find the Factorial. **(A)** 2) Write a stored procedure to find maximum number out of two numbers. **(A)** 3) Write a stored procedure to find square of number. **(B)** 4) Write a stored procedure to find the sum of first 50 even numbers. **(B)** 5) Write a Stored Procedure that returns Total marks of 3 subjects using output parameter if marks of all 3 subject is >35 else return a proper message. **(C)**
8	Function, Stored Procedures, Trigger and Cursor	Lab-8	1) Create a DDL Trigger for CREATE event and display message of successful event. **(A)** 2) Create a DDL Trigger for ALTER event and display message of successful event. **(A)** 3) Create a DDL Trigger for DROP event and display message of successful event. **(A)** 4) Create a DDL Trigger for All three events using log table. **(B)**
9	Function, Stored Procedures, Trigger and Cursor	Lab-9	1) Create a DML After/For Trigger for INSERT event and display the message of the successful event. **(A)** 2) Create a DML After/For Trigger for the UPDATE event and display a message of the successful event. **(A)** 3) Create a DML After/For Trigger for the DELETE event and display the message of the successful event. **(A)** 4) Create a DML After/For Trigger for All three-event using log table. **(B)**

Mrs. Payalben H Hinsu | A.Y.: 2025-26 | Sem./Year: 3 | Course: 2302CS303 - Database Management System - II | Slot Type: Lab (TA) | Division: CE-3A

Sr.	Course Content	Planning Title	Planning Description
10	Function, Stored Procedures, Trigger and Cursor	Lab-10	1) Write T - SQL block to perform Static cursor. **(A)** 2) Write T – SQL block to perform static cursor that fetches first and last value from the result set. **(B)** 3) Write T -SQL block to perform static cursor and try to insert/update/delete a record on table. **(C)**
11	Function, Stored Procedures, Trigger and Cursor	Lab-11	1) Write T - SQL block to perform Dynamic cursor. **(A)** 2) Write T- SQL block to perform Dynamic cursor that fetch value and print that value after some delay. **(B)**
12	Exception Handling, TCC and DCL Commands	Lab-12	1) Write T-SQL block to perform System Defined Exception for data type conversion Exception. **(A)** 2) Write T-SQL block to perform System Defined Exception for divide by 0. **(A)** 3) Write T - SQL to perform User Defined Exception. If User enter age<14 then print the error message "Child labour is not legal". **(A)** 4) Write T-SQL block to generate System Defined Exception for Check constraint violation. **(B)** 5) Write T-SQL block to perform User Defined Exception if we try to delete data that is not available in table. **(B)** 6) Write T – SQL block to generate System Defined Exception for primary key violation. **(C)** 7) Write T-SQL block to multiply three numbers and if any number is 0 then generate User Defined Exception. **(C)**
13	Exception Handling, TCC and DCL Commands	Lab-13	1) Perform queries for SAVEPOINT, ROLLBACK and COMMIT commands on Customer_Master table. **(A)** 2) Perform queries for GRANT and REVOKE command on Employee_Master. **(A)** 3) Perform queries for SAVEPOINT, ROLLBACK and COMMIT commands on Transaction_Master table. **(B)** 4) Perform queries for GRANT and REVOKE command on Account_Master table. **(B)**
14	Normalization and Functional Dependencies	Lab-14	Design Database on any system.
15	Transaction Processing	Lab-15	Implement different types of lock and deadlock.